

APPROVED METHOD:
TOTAL LUMINOUS FLUX
MEASUREMENT OF LAMPS USING AN
INTEGRATING SPHERE PHOTOMETER
AN AMERICAN NATIONAL STANDARD

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has been approved by IES.
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**Prepared by
The IES Testing Procedures Committee**



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CONTENTS

Foreword	1
1.0 Introduction and Scope	1
1.1 Introduction	1
1.2 Scope	1
2.0 Normative References	1
3.0 Nomenclature and Definitions	1
3.1 total luminous flux	1
3.2 sphere efficacy	2
3.3 sphere responsivity	2
4.0 Physical and Environmental Test Conditions	2
5.0 Integrating Sphere Equipment	2
5.1 Integrating Sphere Geometry: 4π	2
5.2 Integrating Sphere Geometry: 2π	3
5.3 Sphere Coating	4
5.4 Baffle	5
5.5 Auxiliary Lamp	6
5.6 Photometer Detection	6
5.7 Spectroradiometer Detection	7
5.8 Lamp Holder and Mounting Hardware	8
5.9 Temperature Sensor	8
5.10 Electrical Circuits	9
6.0 Calibration Standards	9
6.1 Incandescent Standard Lamps	9
6.2 Incandescent Reflector Lamps	9
6.3 Discharge Standard Lamps	9
6.4 Solid-State Lighting Standard Products	10
6.5 Handling of Standards Lamps	10
6.6 Operation of Standard Lamps	10

7.0	Test Procedures	11
7.1	Total Luminous and Total Spectral Spectral Radiant Flux Calibration	11
7.1.1	Integrating Sphere with Photometer.....	11
7.1.2	Integrating Sphere with Spectroradiometer.....	11
7.2	Test Lamp Measurement Procedure	11
7.2.1	Integrating Sphere with Photometer.....	11
7.2.2	Integrating Sphere with SpectroRadiometer	12
7.3	Sources of Error	12
7.3.1	Self-Absorption	12
7.3.2	Spectral Mismatch	13
7.3.3	Sphere Angular Non-uniformity	14
7.3.4	Near-Field Absorption.....	14
7.3.4	Temperature Effects.....	15
8.0	Uncertainty Evaluation	15
9.0	Maintenance of Integrating Spheres	16
	Informative References	16
	Annex A – Basic Integrating Sphere Theory	17
	Annex B – Measurement of the Relative Spectral Throughput of an Integrating Sphere	19
	Annex C – Correction for the Angular Non-uniformity Errors of an Integrating Sphere	19
	Annex D – Alternative Geometries and Calibration Methods	20

Foreword

This document is one of a continuing series of IES Approved Methods prepared to define a baseline for acceptable photometric procedures leading to improved agreement among laboratories. This Approved Method document explains a particular technique for total flux measurement of all types of lamps and luminaires using integrating spheres. The main improvement in this document compared to IES LM-78-2007 is the addition of spectral measurements. While most statements are written using *lamp*, they apply to lamps and luminaires.

1.0 Introduction and Scope

1.1 Introduction

Total luminous flux is the angularly integrated, photopically weighted, total light output from a lamp. Total luminous flux is one of the most important parameters for non-directional, general lighting products. While luminous intensity and beam angle are typically more important for directional lamps, such as reflector lamps, and luminaires, total luminous flux is still an important quantity. Regulations are usually based on luminous efficacy (lm/W), which is the total luminous flux divided by the electrical input power of the light-emitting device. This document describes techniques for measuring the total flux of lamps and luminaires. One method is exclusive to total luminous flux measurements: an integrating sphere system using a photometer. The second method described, an integrating sphere system using a spectroradiometer, is used to measure a variety of angularly integrated quantities. The integrating sphere system has the advantage of fast measurements and does not require a dark room. Air movement is minimized and temperature within the sphere is not subject to the fluctuations potentially present in a temperature-controlled room. The approved method is based on the comparison of the total luminous flux or total spectral radiant flux of a test lamp or luminaire to the total luminous flux or total spectral radiant flux of a standard lamp in an integrating sphere system.

While photometric and spectral integrating sphere measurement systems are very similar, they each have positive and negative aspects. Photometric measurement systems use a $V(\lambda)$ -corrected photometer head, which suffers from spectral mismatch errors. Spectral mismatch errors occur because relative spectral responsivity of the integrating sphere photometric system deviates from the $V(\lambda)$ function. The spectral integrating sphere measurement system does not have spectral mismatch errors. Spectral integrating sphere measurement systems are capable of measuring the integrated color quantities of a light-emitting device; however, they are susceptible to spectral stray light and stability concerns. The two measurement techniques are compared and contrasted in this document along with the techniques required to make high quality measurements.

1.2 Scope

This approved method describes the procedures to be followed and precautions to be observed in performing reproducible measurements of total flux of lamps and luminaires using integrating sphere measurement systems. Two types of integrating sphere systems are presented, one employing a $V(\lambda)$ -corrected photometer head, and another employing a spectroradiometer as the detector. In addition to the specifics of the two measurement systems, the common qualities and measurement techniques are discussed along with calibration and uncertainty analysis considerations.

2.0 Normative References

Illuminating Engineering Society. LS-1-20, Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020.

3.0 Nomenclature and Definitions

3.1 total luminous flux

While the luminous flux (Φ_v) is defined in the International Lighting Vocabulary (ILV; CIE 2011) as the quantity derived from the radiant flux, Φ_r , by evaluating the

radiation according to its action upon the CIE standard photometric observer, the total luminous flux is the cumulative luminous flux of a light emitting device over the solid angle of 4π sr. The total luminous flux is often called "luminous flux" in photometry practice. The unit of total luminous flux is the lumen.

3.2 sphere efficacy

Sphere efficacy (E_d/Φ_v) is the ratio of the illuminance E_d at the detector port to the total luminous flux Φ_v of a light emitting device in the sphere, in units of $1/\text{m}^2$.

3.3 sphere responsivity

The sphere responsivity (y_d/Φ_v) of the system is the ratio of the photometer signal y_d to the total luminous flux Φ_v of a light emitting device in the sphere.

4.0 Physical and Environmental Test Conditions

The integrating sphere efficacy can be slightly affected by changes in temperature or humidity. The responsivity of the photometer head on the integrating sphere is also affected by changes of temperature (typically approximately 0.1 percent per degree Celsius of change). Further integrating sphere efficacy is affected by dust and particulates that accumulate on the integrating sphere coating. Thus, it is preferred to place the integrating sphere in a clean, reasonably temperature-controlled environment. For lighting devices that emit a significant amount of heat, the integrating sphere may be slightly ajar during the stabilization time. Once the lighting device is stabilized, the sphere should be closed and the measurements made as quickly as possible.

The ambient conditions are more critical for measurements of particular types of discharge lamps and solid-state lighting products. For example, fluorescent lamps are required to be operated in an ambient temperature of $25\text{ }^{\circ}\text{C} \pm 1\text{ }^{\circ}\text{C}$. (Refer to the appropriate LM document for the required ambient conditions for each lamp type to be measured.)

5.0 Integrating Sphere Equipment

5.1 Integrating Sphere Geometry: 4π

Figure 1 shows the geometry of a typical integrating sphere system for total flux measurements where the test lamp is mounted in the center of the sphere and the light output is measured in all directions (4π steradians, sr). The integrating sphere is equipped with a photometer head or spectroradiometer, a baffle, an auxiliary lamp assembly, a diffusely reflective sphere coating, lamp mounting hardware, lamp holder and associated electrical wiring.

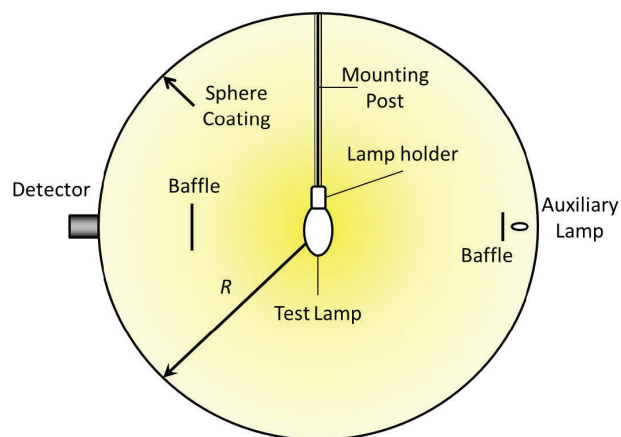


Figure 1: Basic geometry of a 4π integrating sphere measurement system. (Image courtesy of Yoshi Ohno and Cameron Miller.)

The optimum size of the sphere should be chosen by considering the following factors:

1. Size of the Lamp (thus, required size of the baffle)
2. Self-absorption of the lamp, lamp holder and lamp mounting hardware (see **Section 7.3.1**)
3. Internal sphere temperature
4. Sphere responsivity (lowest measurable total flux)

Factors A, B, and C drive the sphere to be as large as possible, while factor D limits the sphere size. Factors B and D are affected by the chosen reflectivity of the sphere coating. In the 4π geometry, as a guideline, the total surface area of the lamp should be less

* This corresponds to, for example, a spherical object less than 30 cm in diameter in a 2-m integrating sphere.

than 2 percent of the total area of the sphere wall.* A general rule is to have the sphere diameter larger than 10 times the *largest dimension* of small lamps and 1.5 times the length of linear lamps to be measured. When linear fluorescent lamps and other linear lamps are measured in an integrating sphere against an incandescent standard lamp, an error occurs due to the difference in geometrical shape (length) of the lamp. If a part of the light source is near the sphere wall, the sphere responsivity for that part of the source will be slightly higher because the emitted light is incident at larger angles than normal, and the coating reflectance increases as the angle of incidence increases (e.g., approximately 0.4 percent increase at 45° incidence for pressed barium sulfate). The angular dependence of the diffuse reflectance of the coating depends on the material and surface conditions. If the size of the sphere is determined as recommended (sphere diameter larger than 1.5 times the length of the linear lamp), this error should be negligible. Commercial integrating spheres of 1 m to 3 m in diameter are used for total flux measurements of lamps.

5.2 Integrating Sphere Geometry: 2π

Figure 2 shows the geometry of an integrating sphere system for total flux measurements where the test lamp is mounted on the side of the sphere and the light output is measured from an emitting plane (2π sr). The integrating sphere is equipped with a photometer head or spectroradiometer, a baffle, an auxiliary lamp assembly, and a diffusely reflective sphere coating. The lamp mounting hardware, lamp holder and associated electrical wiring are outside the sphere.

Similar considerations need to be made for a 2π integrating sphere measurement system as for a 4π system. However, because the associated gear of the lamp, along with most of the heat generated, is outside of the sphere, the sphere does not need to be as large. The 2π geometry is only acceptable for lamps emitting light in forward

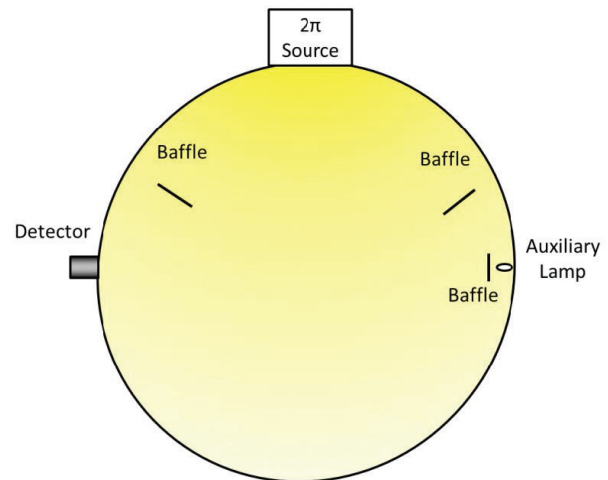


Figure 2. Basic geometry of a 2π integrating sphere measurement system. (Image courtesy of Yoshi Ohno and Cameron Miller.)

directions (regardless of orientation), and may be used for a lamp having a large housing or a luminaire that is too large to use the 4π geometry. The 2π geometry may have considerations that are not present in the 4π geometry.

In the 2π geometry, the opening diameter to mount a lamp should be less than 1/3 of the diameter of the sphere; i.e., the luminaire mount area is less than 3 percent of the integrating sphere surface area. The lamp shall be mounted within the opening and in such a way that its front edges are flush with the edges of the opening, or it can be slightly inside the sphere to ensure that all emitted light is caught in the sphere. The gaps between the edges of the opening and the lamp (or primary standard source) can be covered with a surface (inner side is white) in order that the measurement can be made in a room with normal ambient lighting, as the sphere is completely shielded from outside as shown in **Figure 3**. If this is not convenient and the gaps are to be kept open, a dark room arrangement (around the opening, at least) may be necessary so that no external

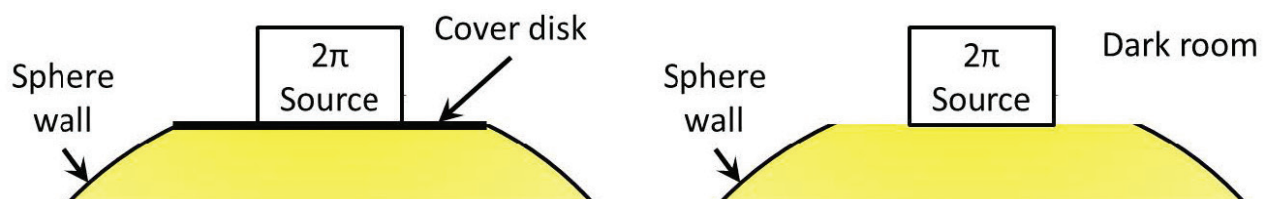


Figure 3. Mounting conditions for a 2π sphere system, using a cover disk (left) and without a cover disk (right). (Image courtesy of Yoshi Ohno and Cameron Miller.)

light or reflected light enters the sphere. In either case, the lamp under test should be mounted to the sphere so that support material or structure will not conduct the heat away from the lamp product.

5.3 Sphere Coating

Sphere coatings should be highly diffusive with appropriate reflectance, and are typically based on barium sulfate. PTFE (polytetrafluoroethylene) is also used in specific applications where high reflectance is required, especially in the ultraviolet region. The reflectance of the coating is an important factor for an integrating sphere and, if necessary, the level of reflectance can be lowered and made more spectrally uniform by mixing light-absorbing material into the high-reflectance white coating material. In the past, the photometers and spectroradiometers were not as sensitive as those currently available; therefore, a high reflectance coating (95 percent to 98 percent) was a better option. In addition, a high reflectance coating is less sensitive to light sources that have angular non-uniformity, such as directional lamps. A useful analysis of an integrating sphere is to measure the sphere angular response distribution function (SRDF). A method of determining the SRDF is to use a beam source that can rotate vertically and horizontally within the sphere, shown in **Figure 4**. As the beam source, which does not change intensity with position, is rotated, the detector signal is measured.

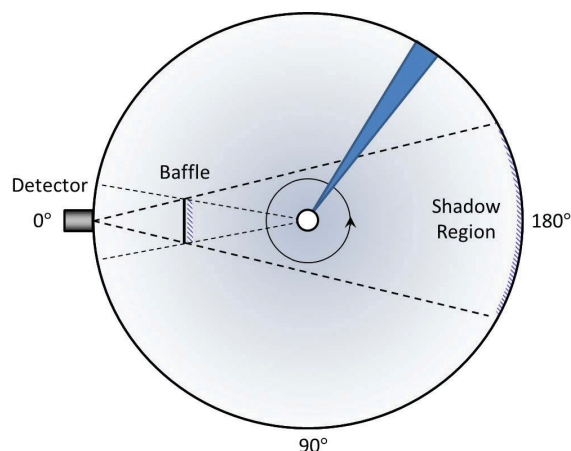


Figure 4. Measurement geometry for determining sphere angular response distribution function (SRDF). (Image courtesy of Yoshi Ohno and Cameron Miller.)

Figure 5 shows a simulation of the SRDF determined for a 2-m sphere with a sphere wall and a 30-cm baffle with a reflectance of 80 percent (no auxiliary lamp).

The sphere responsivity rises as the directional light approaches the edge of baffle and then drops when the initial light hits the baffle. The sphere responsivity also drops significantly when the directional light hits the sphere wall in the shadow region created by the detector baffle (see **Figure 4**). Therefore, any light source that has a non-uniform distribution compared to the primary standard lamp, which has a significant fraction of light illuminating the shadow region or the baffle area, is going to have error in the measurement.

Figure 6 shows a simulation of the SRDF for the same sphere as in **Figure 5**, but the sphere wall and 30-cm baffle have a reflectance of 98 percent. The shadow region and baffle are not nearly as significant. However, with a high reflectance coating, the sphere responsivity is more sensitive to self-absorption by the lamp, contamination by dust, or the reproducibility of opening and closing the sphere. **Annex A** provides a theoretical explanation on the effect of sphere wall reflectance on the performance of the sphere. A quantity derived in **Annex A** is the sphere efficacy, defined as the illuminance on the sphere wall divided by the light source flux:

$$\frac{E_d}{\Phi_v} = \frac{\rho}{1 - \rho} \times \frac{1}{4\pi \cdot R^2}, \quad (1)$$

where:

ρ = reflectance of the sphere wall and

R = radius of the integrating sphere.

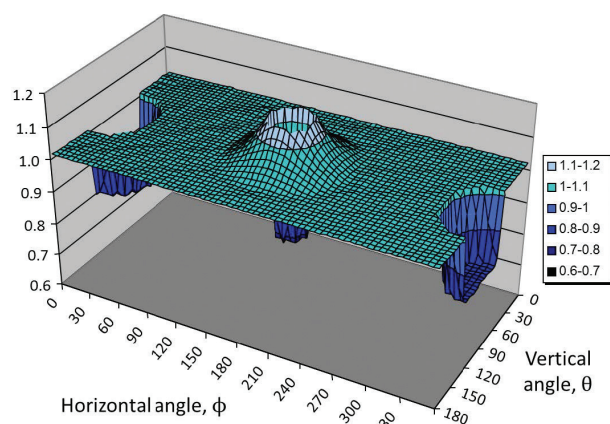


Figure 5. SRDF for a 2-meter sphere, where the sphere wall and 30-cm baffle have a reflectance of 80 percent and there is no auxiliary lamp. (Image courtesy of Yoshi Ohno and Cameron Miller.)

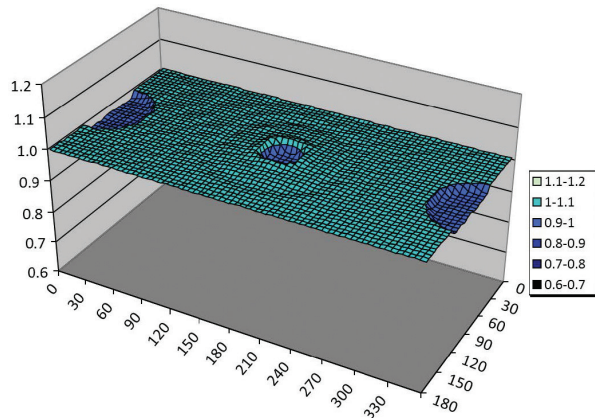


Figure 6. SRDF for a 2-meter sphere, where the sphere wall and a 30-cm baffle have a reflectance of 98 percent and there is no auxiliary lamp. (Image courtesy of Yoshi Ohno and Cameron Miller.)

As approaches 100 percent, the denominator becomes very small. If, for example, changes in sphere reflectance of 0.05 percent occur due to opening and reclosing the sphere, the sphere efficacy changes by 2.5 percent if the sphere wall reflectance is 98 percent. If the sphere wall reflectance is 80 percent, the sphere efficacy only changes by 0.3 percent. Furthermore, a sphere with a lower reflectance is more stable over time, perhaps making it a better choice for an integrating sphere system that is only calibrated occasionally. Depending on the sphere surface type, humidity can cause a similar effect.

A concern that has developed with solid-state lighting (SSL) that needs to be addressed in the measurement procedures is the use of phosphors as wavelength converters. To achieve a certain correlated color temperature, many SSL products rely on blue light emitting diodes (LEDs) to excite a phosphor that emits in the yellow region of the spectrum. The combination of the transmitted blue light and yellow light achieves the desired correlated color temperature. In an integrating sphere, the blue light that is emitted has the opportunity to be reflected back to the SSL product to be absorbed and re-emitted as yellow light. This effect is going to be more significant in smaller integrating spheres with sphere coatings that have a high reflectivity. Traditional linear fluorescent tubes, which use mercury excited-atom emission at 254 nm to excite the phosphors, do not have an issue because little 254-nm light leaves the

lamp—any such emission that is not absorbed by the phosphor is absorbed by the glass.

For a commercial testing laboratory using broadband photopic head detectors, a sphere coating that has a reflectance of 80 percent to 90 percent is recommended. For laboratories using spectral detectors, a coating that has a reflectance of 90 percent to 98 percent is recommended.

5.4 Baffle

A baffle is needed to shield the detector from direct illumination by the lamp (see **Annex A**). However, the baffle is the main cause of the angular non-uniformity of the sphere responsivity. In order to minimize the angular non-uniformity errors, the baffle should be located a distance of $\frac{1}{3}$ to $\frac{1}{2}$ of the sphere radius from the detector. The size of the baffle should be made the smallest possible to shield the detector from direct illumination from the standard and test lamps (see **Figure 7**). The surface of the baffle facing the lamp should have the highest reflectance possible. The surface reflectance of the baffle facing the detector is not critical and can be coated with the same coating as the sphere wall.

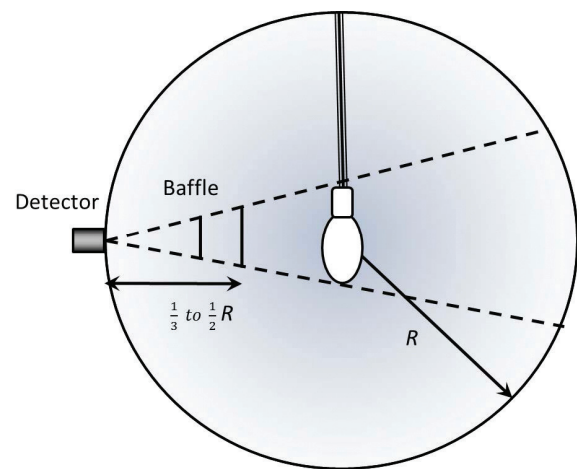


Figure 7. Baffle size and location. (Image courtesy of Yoshi Ohno and Cameron Miller.)

For linear lamps, the shape of the baffle depends on the orientation of the lamp, as shown in **Figure 8**. The coaxial arrangement is advantageous in keeping the baffle size small. However, mounting arrangements for linear lamps, as well as sphere access, make the perpendicular

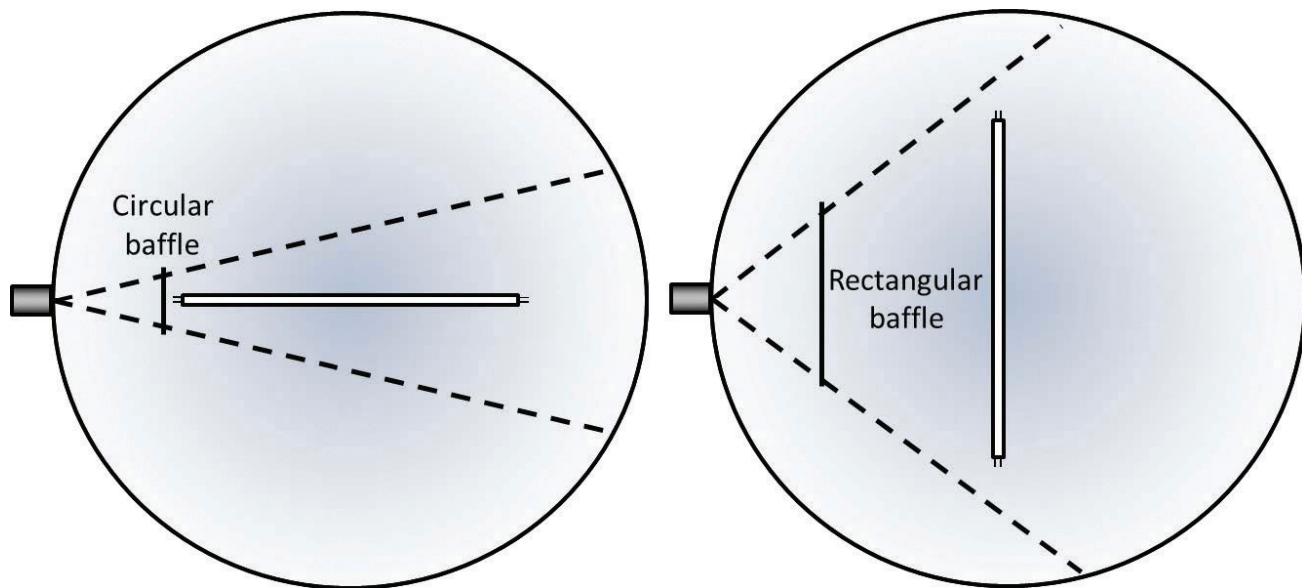


Figure 8. Arrangements for linear lamps. Left: coaxial arrangement. Right: perpendicular arrangement. (Image courtesy of Yoshi Ohno and Cameron Miller.)

arrangement preferable in many practical situations. In this case, the baffle is a rectangular shape, possibly with a larger section in the middle to effectively shield both the fluorescent lamp and incandescent standard lamp.

5.5 Auxiliary Lamp

Integrating sphere measurement systems should be equipped with an auxiliary lamp to allow measurement of self-absorption (see **Section 7.3.1**) by lamps and associated hardware. The auxiliary lamp provides illumination to the entire sphere wall, but it should be shielded from the detector and the lamp to be measured. The auxiliary lamp shall be broadband, covering the entire spectral region of interest. An auxiliary lamp is preferably placed on the sphere wall opposite from the detector, but it can be placed elsewhere. The auxiliary lamp and its baffle should be small enough not to affect the performance of the integrating sphere (typically less than 5 percent of the diameter of the sphere). The self-absorption measurement is normally carried out by alternately positioning the standard and test lamps in the same location in the sphere, which requires repeated opening and closing of the sphere, while the auxiliary lamp is operating. Therefore, the auxiliary lamp should be stable against shocks resulting from opening and closing the sphere. Typically, an auxiliary lamp would be a low-voltage quartz tungsten-halogen lamp, as such a lamp meets all the above

requirements. Monitoring the light output of a lamp operating at a controlled constant current can provide a good indication of the stability of the integrating sphere system. For an integrating sphere measurement system using a photometer, it is advantageous to use an auxiliary lamp that has a spectral distribution similar to that of the test lamp measured, so that self-absorption is measured accurately. This is especially important when the self-absorption is very large ($\alpha < 0.85$; see **Section 7.3.1**) or when the housing of a luminaire under test is large and strongly colored.

5.6 Photometer Detection

Integrating sphere measurement systems with a photometer as the detection device should use a $V(\lambda)$ -corrected photometer that accounts for the multiple reflections of the sphere coating and the transmittance of an external diffuser (if present). As a guideline for the degree of the spectral match to the $V(\lambda)$ function, it is preferable that the value of the photometer be less than 3 percent (see **Section 7.3.2**). When combined with the spectral variation of the paint, the value is usually much higher especially for high-reflectance coatings that tend to have significantly lower reflectance in the blue region of the spectrum. For high accuracy measurements, spectral mismatch corrections are required. (Refer to **Section 7.3** for the determination of the spectral mismatch correction methodology.)

In addition, the photometer should have a cosine-corrected angular responsivity according to the theory of an integrating sphere (see **Annex A**). Cosine response can be achieved by using a light-transmitting diffuser in front of the photometer head, as employed in illuminance meters. A small satellite sphere can also be used for cosine correction. While the accuracy of the cosine correction is not critical, it is important that the photometer be mounted at a detector port of the sphere, with its reference plane flush with the coating surface so that the photometer receives light from all of the sphere wall surfaces. The cosine response of the photometer should have a directional response index of less than 15 percent. It is poor practice to arrange the photometer head to view only the backside of the baffle by recessing the photometer head in the sphere wall or not having a diffuser. This would cause large angular non-uniformity of the sphere responsivity (refer to **Section 7.3.3**) and could lead to serious errors in substitution measurements of lamps having different angular luminous intensity distributions than the reference standard.

The responsivity of a photometer head can drift with variations of ambient temperature, or due to heat from the lamp mounted in the sphere. A temperature-controlled or temperature-monitored photometer head should be employed. Temperature control or monitoring should be applied to the part of the photometer head that includes the $V(\lambda)$ filter and photodiode. Temperature-monitored photometers should apply correction factors to achieve lower uncertainties.

A well-designed photometer employing a high quality silicon photodiode can have a linear response over eight orders of magnitude. Detectors other than silicon photodiodes are not recommended. To measure the photocurrent generated in the photometer, a current-to-voltage (transimpedance) amplifier should be used. An amplifier built into the photometer is preferred if low noise and a wide dynamic range are required. Together with the wide linearity range of a silicon photodiode, a low-noise amplifier with several gain ranges (e.g., 10^4 to 10^8 V/A) allows luminous flux measurements over several orders of magnitude using one integrating sphere. A picoammeter may be adequate for reading the signal, negating the need for a current-to-voltage

amplifier. Solid-state lighting products may have a rapidly oscillating optical signal (ranging from 120 Hz to greater than 85 kHz). While the silicon photodiode response time is fast enough to capture the optical signal, current-to-voltage amplifiers may be time-constant limited. A fast response, high quality voltmeter is required to obtain accurate results. A slow response, low quality voltmeter may give erroneous results without any warnings. In any case, the voltmeter used for measurement of a photometer signal shall always be set to DC mode, regardless of whether it is a DC signal or a DC signal plus rapid-changing high-frequency AC signals, in order to obtain the mean voltage signal. Another option is to use a charge-capturing electronic device that integrates the photocurrent over time and then is read using an electrometer.

5.7 Spectroradiometer Detection

Either mechanical scanning or array-type spectroradiometers may be used. The array spectroradiometer has the advantage of shorter measurement time due to the multiplex nature of arrays. While the defined visible spectral region ranges from 360 nm to 830 nm, the spectroradiometer shall have a minimum spectral range from 380 nm to 780 nm, which is sufficient for lumen and color information.

The detector port of the integrating sphere, similar to that for the photometer, shall be a flat diffuser or a satellite sphere mounted flush to the sphere coating surface, so that the input of the spectroradiometer at the detector port has an approximate cosine response, with the directional response index being less than 15 percent. It should be noted that an optical fiber input (with no additional optics), often provided with array spectrometers, has a narrow acceptance angle and is not acceptable unless it is used with additional optics for cosine correction.

Calibrated spectroradiometers measure photometric quantities without spectral mismatch errors; however, there remain many other sources of error associated with spectroradiometers. Note that errors in some low quality array spectroradiometers can be larger than those from high quality photometer heads. Errors can be significant when the spectral distribution of a test lamp is dissimilar to the standard source. Major sources of

error include bandwidth, scanning interval, wavelength accuracy, spectral stray light, detector nonlinearity, and input geometry. For accurate colorimetry, bandwidth and scanning intervals of 5 nm or smaller are required for spectroradiometric methods. Other recommendations given in LM-58-13 should be followed to minimize the errors and evaluate measurement uncertainties.

A sampling time of 1 millisecond (ms) is the typical lower limit for most charge-coupled device (CCD)-based spectroradiometers. The shorter sampling time means faster online device test processing, as well as greater dynamic range for measuring high flux of sources. The sampling time has two components: the integration time for one scan, and the number of scans that are averaged together. For many CCD-based spectroradiometers, the time spent reading the CCD is very small compared to the integration time. The integration time shall be chosen to ensure that the detector does not become saturated. In addition, the sampling time needs to be long enough to ensure that for devices not having a constant optical output (such as flicker or pulse-width modulated sources) several cycles are sampled. This can be accomplished by ensuring that the integration time, or the integration time averaged over the number of scans, is an integer multiple of the cycle time. As an example for a light emitting device that has a 60-Hz component, the integration time, or the integration time averaged over the number of scans, should be an integer multiple of 16.7 ms. The benefit of averaging scans is increasing the signal-to-noise ratio. For smaller integrating sphere systems with spectroradiometers, the integration time may be smaller than the cycle-time interval. Another option in this case is to calibrate the spectroradiometer with a neutral density filter in the optical path, which will increase the integration time.

5.8 Lamp Holder and Mounting Hardware

Integrating sphere photometers are often equipped with interchangeable mounting hardware and lamp holders to accommodate different

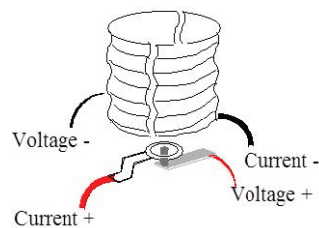


Figure 9. Schematic depiction of a four-pole socket.
(Image courtesy of Yoshi Ohno and Cameron Miller.)

lamp types. The external surfaces of the lamp holder and mounting hardware should be painted with a high-reflectance coating (e.g., barium sulfate paint with 98 percent reflectance). For the highest-accuracy measurement, it is desirable to use a four-pole contact lamp holder (Kelvin-type arrangement for an Edison-based lamp, as shown in **Figure 9**). This type of lamp holder allows measurement of lamp voltage that is not affected by voltage drops at the contacts where the current is supplied. By monitoring the lamp voltage at constant current, changes of standard lamps caused by transportation, aging, or other unknown reasons, can be detected.

Any dark surfaces or gaps on the lamp holder can cause near-field absorption errors (see **Section 7.3.4**). If there is a large gap between the lamp holder and lamp base, it may cause significant errors with small frosted-bulb lamps. Special care should be taken when measuring miniature lamps because the size of the lamp holder tends to be much larger than the size of the lamp, and the absorption of light by the lamp holder tends to be significant. To avoid this problem, a miniature lamp can be calibrated with its designated lamp holder, where the combination of the lamp and lamp holder is regarded as a lamp and never detached once calibrated.

5.9 Temperature Sensor

The ambient temperature in the integrating sphere should be monitored with a temperature sensor installed in the sphere. The location of the sensor, if used, shall be at a point in the sphere not more than 1 m from the lamp and at the same height as the lamp. In addition, the temperature sensor shall be shielded from the direct lamp radiation and radiation from any other source. It is recommended to monitor the temperature inside the sphere for low-pressure discharge lamps and solid-state lighting products, many of which are sensitive to ambient temperature. A lamp mounted on the sphere wall requires the ambient temperature to be measured behind the baffle in the sphere, in addition to the ambient air outside the sphere near the lamp. Individual IES laboratory methods have specifications for acceptable temperature limits. Monitoring the temperature inside the sphere also provides some indication of the possible temperature effects on the photometer head (see **Section 5.6**) and on the sphere coating (see **Section 5.3**).

5.10 Electrical Circuits

The operating circuits for the test lamps vary by type. Other IES laboratory methods should be referred to for the specifications of operating circuits for each type of lamp and the requirements for the power supplies and other electrical equipment used.

A general concern is the transmittance of short-pulse electricity (< 0.1 ms) or high-frequency electricity (> 10 kHz). Long lengths of parallel wires can have significant inductive and capacitive impedance, which can alter the current or voltage readings at high frequency, as do capacitive coupled ground loops if not properly grounded. For large capacitive impedance effects, the power analyzer may be a larger current draw than the light emitting device, causing errors in the electrical measurements. If measurements are to be done when high-frequency components are carried by the wires within an integrating sphere, coaxial cables should be used or the wires separated or twisted, depending on the problem to be solved.

6.0 Calibration Standards

Total luminous flux and total spectral radiant flux standard lamps (primary standards, working standards, and/or transfer standards) should be reproducible and stable for repeated and long-term use. Only limited types of lamps are suitable for use as total luminous and total spectral radiant flux standards. The lamps should be recalibrated at an interval specified in terms of total operating time, which depends on the aging rate of the lamps. It is recommended that a group of three or more standard lamps be maintained as primary standards for a laboratory and that their scale be transferred to working standard lamps for routine calibration work.

Before new total luminous or total spectral radiant flux standard lamps are calibrated, the lamps should be seasoned and tested for stability. (Refer to IES LM-54-13 for seasoning specifics for different types of lamps.)

6.1 Incandescent Standard Lamps

Calibration services for incandescent lamps, including

halogen lamps, for the quantities of total luminous flux or total spectral radiant flux are available from national metrology laboratories and commercial calibration laboratories, many of which are accredited calibration laboratories. The DC operating current, electrical polarity and operating position should be specified for each lamp to be calibrated. The operating current of lamps calibrated for total luminous flux is often set for a correlated color temperature (CCT) of approximately 2856 K. The operating current of quartz tungsten-halogen lamps for total spectral radiant flux is often set for a CCT of 3100 K to 3200 K to ensure as much blue light as possible.

For some types of total luminous flux lamps, lower CCTs (2700 K to 2800 K) are used to reduce their aging rate. The aging rate of lamps is an important factor in determining the calibration cycle of the standard lamps. The aging rate of normal gas-filled incandescent standard lamps can be as high as 1 percent per 24-hour operation at 2856 K. Quartz tungsten-halogen lamps, particularly those designed for lower voltage, e.g., 28 volts, generally have a significantly lower aging rate that allows them to be operated at higher temperature. Incandescent lamps, including halogen lamps, are sensitive to their operating electrical condition and position. The luminous flux changes if their operating condition is changed.

6.2 Incandescent Reflector Lamps

Since integrating spheres can have angular non-uniformity, particularly between the top and bottom portions of the sphere, standard incandescent reflector lamps can be used to estimate the degree of angular non-uniformity. By calibrating a 4π sphere system with an isotropic incandescent standard lamp and then measuring a calibrated standard incandescent reflector lamp, the difference will provide an estimation of the sphere angular non-uniformity when measuring directional lamps. Another application of incandescent reflector lamps is calibration of a sphere used in the 2π configuration.

6.3 Discharge Standard Lamps

Discharge lamps, such as linear fluorescent lamps and some high intensity discharge (HID) lamps, selected for acceptable stability and reproducibility can be

used as standards for limited purposes. Discharge lamps, however, are generally not as reproducible as incandescent standard lamps, and they are often used as check standards rather than primary standards to calibrate integrating spheres. If these types of lamps are to be used as primary standard lamps, the uncertainty should be carefully evaluated based on the variation in total luminous flux values in several repeated measurements. Some lamps will have storage effects that should be accounted for in the uncertainty analysis. (Refer to IES laboratory methods for particular discharge lamps guidance.)

Most discharge lamps are not acceptable candidates for total spectral radiant flux standard lamps due to their sharp spectral features. However, the sharp spectral feature of low-pressure mercury discharge lamps makes them excellent artifacts to calibrate or check calibration of spectral detectors with respect to the wavelength scale.

6.4 Solid-State Lighting Standard Products

Solid-state lighting products are not widely used as standard lamps. Solid-state lighting products are temperature sensitive; therefore, special temperature controlling gear is required to ensure lumen stability. While solid-state lighting products have a smoother spectral distribution than discharge lamps, the spectral shape still has significant slopes, causing certain regions of the spectrum to have increased uncertainties. This is due to the required level of uncertainty for the wavelength calibration of either a spectroradiometer detector or the spectral responsivity of the sphere measurement system using a photometer. However, the use of stable and reproducible solid-state lighting (SSL) products as primary standard lamps is advantageous with regard to reducing spectral mismatch errors in a photometer system or cancelling the wavelength calibration issue, if the spectral distribution is very similar to the test lamp to be measured.

6.5 Handling of Standard Lamps

Standard lamps should be handled carefully to avoid mechanical shocks, and the lamps should be handled using gloves to avoid contamination by fingerprints. The lamp identification number should be permanently marked on each lamp. Prior to calibration, the bulbs of

the lamps should be cleaned with a soft, lint-free cloth to remove dust from packing material. After operation of the lamps, they should be allowed to cool before removal from the socket, as some type of lamps get very hot during operation. For incandescent lamps, the cool-down period should be at least 5 minutes. The filament is very fragile during operation, so mechanical shock of the lamp should be avoided. Standard lamps should be kept in a closed container when not in use.

6.6 Operation of Standard Lamps

As mentioned above, incandescent, including halogen, standard lamps are operated at a specified lamp current, which is often set to give a specific correlated color temperature (e.g., 2856 K). The lamp current is ramped up and down slowly (typically over 30 s) to avoid high inrush currents and associated localized thermal shock to the filaments. Photometric measurements should be made after the lamp has stabilized, which normally takes 5 to 10 minutes. Lamp current, voltage and light output can be monitored to determine when a required or sufficient level of stability has been achieved. The lamp should not be operated longer than necessary, in order to keep the total operating time of the lamp to a minimum. A record of the operating time of each standard lamp shall be kept.

Linear fluorescent lamps are operated according to specifications referenced in LM-9-09, except the lamps are operated at a set value of the current without cathode heat. The integrating sphere can be opened during lamp stabilization, if necessary, to maintain the required ambient temperature for the measurement. In this case, before the measurement starts, the sphere should be closed slowly to avoid cooling the lamp by air movement or disturbing the mercury at the cold spot.

Currently available SSL standard products are based on operating with fixed direct current along with temperature stabilizing gear. Well-engineered SSL standard products can be stabilized in a few tens of seconds, but have long-term instabilities if not used regularly.

For measurements of incandescent and various discharge test lamps, refer to the appropriate IES laboratory methods for their operation requirements.

7.0 Test Procedures

The total luminous flux or total spectral radiant flux of a test lamp shall be determined by direct comparison to the primary standard lamp that is used to calibrate the integrating sphere system. In addition to the total luminous flux or the total spectral radiant flux measurement, the self-absorption measurements (see **Section 7.3.1**) and other correction measurements (see **Section 7.3**) should be conducted as necessary.

7.1 Total Luminous and Total Spectral Radiant Flux Calibration

7.1.1 Integrating Sphere with Photometer. Using a photometer as the detector for an integrating sphere is the more traditional approach of integrating sphere photometry. This method is acceptable but less preferred than using a spectroradiometer, due to the potential for large spectral mismatch errors in the measurement of the total luminous flux of the test lamp (if the spectral mismatch corrections are not applied), and due to a need for a separate measurement instrument for color or color-rendering quantities.

At the beginning of a calibration cycle, each standard lamp is operated in the integrating sphere measurement system. After the lamp is stabilized, the photometer signal and lamp electrical data shall be recorded. The sphere responsivity, $R_{v,s}$ [A/lm], for each standard lamp having luminous flux [lm] is obtained by

$$R_{v,s} = \frac{y_s - y_{Dark,s}}{\Phi_{v,s}} \cdot k_{s,s}, \quad (2)$$

where:

- y_s = the photometer signal with the standard lamp on
- $y_{Dark,s}$ = the photometer signal with the standard lamp off
- $y_{s,s}$ = the sphere angular non-uniformity correction factor for the standard lamp with respect to an isotropic point source (see **Annex C**)

If $k_{s,s}$ is not determined, $k_{s,s} = 1$ should be used, and the resulting uncertainty should be considered. For better

accuracy, the calibration process may be repeated at the end of the measurement of test lamps, and the average value of before and after the test lamp measurement may be used. Check standards should be used to verify the calibration as well as show any drift between calibrations. A good time for a new calibration is when the check lamp change exceeds a certain amount, e.g., 1.0 percent, which is dependent on the desired level of uncertainty.

7.1.2 Integrating Sphere with Spectroradiometer.

An integrating sphere measurement system with a spectroradiometer as a detector measures total spectral radiant flux (unit: W/nm), from which total luminous flux and color quantities are calculated. By using an array spectroradiometer, the measurement speed can be of the same magnitude as for a photometer.

The instrument (integrating sphere plus spectroradiometer) shall be calibrated against a primary standard calibrated for total spectral radiant flux. Since the integrating sphere is included in this calibration, the spectral throughput of the sphere need not be known. The sphere responsivity, $R_{e,s}(\lambda)$ [A/(W/nm)], for each standard lamp having total spectral radiant flux $\Phi_{e,s}(\lambda)$ [W/nm] is obtained by

$$R_{e,s}(\lambda) = \frac{y_s(\lambda) - y_{Dark,s}(\lambda)}{\Phi_{e,s}(\lambda)} \cdot k_{s,s}, \quad (3)$$

where:

- $y_s(\lambda)$ = the spectroradiometer signal
- $y_{Dark,s}(\lambda)$ = the dark/stray light spectroradiometer signal

It should be noted that the sphere angular non-uniformity correction factor generally does not depend significantly on the wavelength. In order to assess the change in conditions, the calibration process may be repeated at the end of the measurement of test lamps, and the average value of $R_{e,s}(\lambda)$ before and after the test-lamp measurement may be used. In addition, as described in the section above, check standards can be used to verify any drift between calibrations.

7.2 Test lamp Measurement Procedure

7.2.1 Integrating Sphere with Photometer. A test lamp is mounted in the integrating sphere and is operated until it has stabilized. After the test lamp has

stabilized, the photometer signal and lamp electrical data shall be recorded. The total luminous flux of the test lamp $\Phi_{v,t}$ is determined by

$$\Phi_{v,t} = \frac{y_t - y_{Dark,t}}{R_s} \cdot \frac{F}{\alpha} \cdot k_{s,t} , \quad (4)$$

where:

- y_t = the photometer signal for the lamp under test
- $y_{Dark,t}$ = the dark/stray light signal for the lamp under test
- F = the spectral mismatch correction factor (see **Section 7.3.2**)
- α = the self-absorption factor (see **Section 7.3.1**)
- $k_{s,t}$ = the sphere angular non-uniformity correction factor for the test lamp with respect to an isotropic point source (see **Annex C**)

If factor F is not determined, $F = 1$ should be used, and the resulting uncertainty should be considered. Similarly, if $k_{s,t}$ is unknown, $k_{s,t} = 1$ should be used, and the resulting uncertainty should be considered. Substituting Eq. 4 into Eq. 2 gives:

$$\Phi_{v,t} = \Phi_{v,s} \cdot \frac{y_t - y_{Dark,t}}{y_s - y_{Dark,s}} \cdot \frac{F}{\alpha} \cdot \frac{k_{s,t}}{k_{s,s}} , \quad (5)$$

This demonstrates the direct comparison to the primary standard. The sphere angular non-uniformity errors cancel if angular intensity distributions of the test lamp and standard lamp are identical and both lamps are aligned in the sphere in the same way. When the photometer signal is measured, an average of several readings can be recorded together with the standard deviation. The standard deviation indicates the stability of the lamp and power supply in operation and includes detector noise, particularly when measuring sources having very low flux levels.

7.2.2 Integrating Sphere with Spectroradiometer.

To measure the total flux or color quantities, a test lamp is mounted in the integrating sphere and is operated until it has stabilized. After the test lamp has stabilized, the spectroradiometer signal and lamp electrical data shall be recorded. The total spectral radiant flux of the test lamp $\Phi_{e,t}(\lambda)$ is determined by

$$\Phi_{e,t}(\lambda) = \frac{y_t(\lambda) - y_{Dark,t}(\lambda)}{R_{e,s}(\lambda)} \cdot \frac{1}{\alpha(\lambda)} \cdot k_{s,t} , \quad (6)$$

where:

- $y_t(\lambda)$ = the photometer signal for the lamp under test
- $y_{Dark,t}(\lambda)$ = the dark/stray light signal for the lamp under test
- α = the self-absorption factor (see **Section 7.3.1**).

Many array spectroradiometers have software that automatically includes the dark signal subtraction. Substituting Eq. 6 into Eq. 3 gives:

$$\Phi_{e,t}(\lambda) = \Phi_{e,s}(\lambda) \cdot \frac{y_t(\lambda) - y_{Dark,t}(\lambda)}{y_s(\lambda) - y_{Dark,s}(\lambda)} \cdot \frac{1}{\alpha(\lambda)} \cdot \frac{k_{s,t}}{k_{s,s}} , \quad (7)$$

This demonstrates the direct comparison to the primary standard. Similar to an integrating sphere with a photometer, the sphere angular non-uniformity errors cancel if angular intensity distributions of the test lamp and standard lamp are identical and both lamps are aligned in the sphere in the same way. The total radiant flux of the test lamp $\Phi_{e,t}$ is obtained by:

$$\Phi_{e,t} = \int_{\lambda} \Phi_{e,t}(\lambda) d\lambda , \quad (8)$$

The total luminous flux of the test lamp $\Phi_{v,t}$ is obtained by:

$$\Phi_{v,t} = K_m \int_{\lambda} \Phi_{e,t}(\lambda) \cdot V(\lambda) d\lambda , \quad (9)$$

This is derived from the measured total spectral radiant flux $\Phi_{e,t}(\lambda)$ [W/nm], where $K_m = 683$ lm/W.

The total photon flux of the test lamp $\Phi_{p,t}$ is obtained by:

$$\Phi_{p,t} = \int_{\lambda} \Phi_{e,t}(\lambda) \cdot \frac{\lambda}{hc_0} d\lambda , \quad (10)$$

This is derived from the measured total spectral radiant flux $\Phi_{e,t}(\lambda)$ [W/nm], where h is Planck's constant and c_0 is the speed of light in vacuum.

7.3 Sources of Error

7.3.1 Self-Absorption. A lamp mounted in the sphere absorbs some of the light reflected from the sphere wall, thus reducing the integrating sphere photometer responsivity. This effect is called *self-absorption*. The larger the lamp, relative to the size of the sphere, and

the darker the color of the lamp bulb or the more structure in the heat sink for a solid-state lighting device, the more self-absorption the lamp exhibits. The recommended method for correcting this error relies upon use of an auxiliary lamp to determine the magnitude of self-absorption errors. The auxiliary lamp is turned on and allowed to stabilize. The photometer signal y_0 , or the signal from the spectroradiometer $y_0(\lambda)$, is taken for the baseline condition where the lamp (and lamp holder if applicable) is removed from the sphere. The photometer signal y_i , or the signal from the spectroradiometer $y_0(\lambda)$, is taken with each standard and test lamp 1,...i,...n (with an appropriate lamp holder as needed) mounted in the sphere, while the auxiliary lamp is continuously operated at constant lumen output. The test lamp shall not be operated during the self-absorption measurement.

The self-absorption factor, which is a unit-less number typically less than one, depends on the reflectance of the sphere coating; the larger the reflectance, the larger the absorption factor. For the highest-accuracy measurement, the self-absorption of each individual test lamp, even of the same type, may be measured. If lower accuracy is acceptable and documented, the same self-absorption factors can be taken for the same type of lamp, used with the same lamp holder in the same sphere, for a certain period.

The self-absorption factor for an integrating sphere system that uses a photometer can be determined as a ratio of the signals:

$$\alpha = \frac{y_{aux,t}}{y_{aux,s}}, \quad (9)$$

Where, for 4π or 2π geometry:

$y_{aux,t}$ = the photometer signal with the auxiliary lamp on and the lamp under test mounted in the sphere but turned off

$y_{aux,s}$ = the photometer signal for the auxiliary lamp turn on when the total flux primary standard lamp is mounted in the sphere and turned off

The auxiliary lamp can be an incandescent lamp, including halogen, or a white LED source similar to solid-state lighting products under test.

The self-absorption correction is critical, since the physical size and shape of the SSL products under test are typically very different from the primary standard lamp's size and shape. The self-absorption is wavelength dependent because the spectral reflectance of the lamp, lamp holder and sphere coating are not necessarily spectrally flat. The self-absorption factor at a given wavelength is given by:

$$\alpha(\lambda) = \frac{y_{aux,t}(\lambda)}{y_{aux,s}(\lambda)}, \quad (10)$$

where $y_{aux,t}(\lambda)$ and $y_{aux,s}(\lambda)$ are the spectroradiometer readings for the auxiliary lamp when the SSL product under test or the primary total spectral radiant standard are mounted in or on the sphere (4π or 2π geometry). In this case, the SSL product and the primary standard are not operated. Only the auxiliary lamp is operated.

Self-absorption is less than one unless there is a significant amount of fluorescent material exposed on the test lamp or the test lamp is less absorptive than the standard lamp. Emission from fluorescent or phosphorescent materials may result in self-absorption factors greater than one because the blue light from the auxiliary lamp is down-converted to yellow light. The yellow light has a significantly larger photopic weighting than the blue light. This may be significant for products that include remote phosphor components and measurement sphere systems with high reflectivity and small diameter. Theoretically, fluorescent or phosphorescent emission can be compensated for when a spectroradiometer is used (if the auxiliary lamp has a similar spectral power distribution as the source), but it is virtually impossible to correct for fluorescent or phosphorescent emissions using a photometer (Keppens et al.).

7.3.2 Spectral Mismatch. The spectral responsivity of the integrating sphere photometer cannot be perfectly matched to the $V(\lambda)$ function. An error occurs when a test lamp has a different spectral power distribution from that of the standard source. The f'_1 value [ISO/CIE 19476:2014(E)] gives a rough indication of the magnitude of errors that can occur for general white light sources, but errors can be larger than the f'_1 value for test lamps with a few narrowband emissions or for monochromatic sources.

To determine the f'_1 value, the relative spectral responsivity of the total sphere system $S_{rel}(\lambda)$ is required to be determined, which is given as the product of the relative spectral responsivity of the photometer $S_{ph,rel}(\lambda)$ and the relative spectral throughput of the sphere $T_{rel}(\lambda)$:

$$S_{rel}(\lambda) = S_{ph,rel}(\lambda) \cdot T_{rel}(\lambda), \quad (11)$$

The relative spectral responsivity of the photometer $S_{ph,rel}(\lambda)$, should be measured under hemispherical illumination geometry. If it is measured only in a normal direction, the uncertainty contribution should be determined. $T_{rel}(\lambda)$ is theoretically given by:

$$T_{rel}(\lambda) = k \cdot \frac{\rho_a(\lambda)}{1 - \rho_a(\lambda)}, \quad (12)$$

where:

$\rho_a(\lambda)$ = the average reflectance of the entire inner sphere surfaces (including $\rho = 0$ for an opening, if there is one)

k = a normalization factor

If $\rho_a(\lambda)$ of the integrating sphere in use is measured accurately, then $T_{rel}(\lambda)$ may be obtained using this equation. However, integrating spheres in use are contaminated to some degree, and the data of coating samples tend to deviate from the reflectance of the real sphere surface. Therefore, it is recommended that $T_{rel}(\lambda)$ be directly measured on the integrating sphere using the procedures given in **Annex B**.

Once $S_{rel}(\lambda)$ is determined, f'_1 is calculated by:

$$f'_1 = \frac{\int_{\lambda} |S_{rel}(\lambda) - V(\lambda)| d\lambda}{\int_{\lambda} V(\lambda) d\lambda}, \quad (13)$$

with:

$$S_{rel}^*(\lambda) = \frac{\int_{\lambda} S_A(\lambda) V(\lambda) d\lambda}{\int_{\lambda} S_A(\lambda) S_{rel}(\lambda) d\lambda} \cdot S_{rel}(\lambda), \quad (14)$$

where:

$S_A(\lambda)$ is spectral distribution of CIE Illuminant (λ) is the spectral luminous efficiency function

With knowledge of $S_{rel}(\lambda)$ and the relative spectral power distribution $S_t(\lambda)$ of the lamp under test, the spectral mismatch correction factor F is given by:

$$F = \frac{\int_{\lambda} S_s(\lambda) S_{rel}(\lambda) d\lambda \int_{\lambda} S_t(\lambda) V(\lambda) d\lambda}{\int_{\lambda} S_s(\lambda) V(\lambda) d\lambda \int_{\lambda} S_t(\lambda) S_{rel}(\lambda) d\lambda}, \quad (15)$$

Where $S_s(\lambda)$ is the spectral distribution of the reference standard source.

Spectral mismatch errors can be corrected by multiplying the measured lumen value of the test lamp by the correction factor F . The accuracy $S_t(\lambda)$ of is generally not very critical; thus, the nominal spectral distribution of a product may be used.

7.3.3 Sphere Angular Non-uniformity. The responsivity of an integrating sphere is not angularly uniform over the sphere wall due to the effects of the baffle and other objects in the sphere including contamination of the sphere wall and gaps between hemispheres. An error occurs when a test lamp is compared with a standard lamp that has a different angular luminous intensity distribution from that of the test lamp. The magnitude of the error depends strongly on the reflectance of the sphere coating. As an example, the angular non-uniformity error from the intensity distribution of various incandescent standard lamps, as compared to a given incandescent standard lamp, in a 2-m integrating sphere with a 30-cm (diameter) baffle 50 cm from the sphere wall is estimated to be on the order of 0.1 percent (with a 98 percent reflectance coating) to 1 percent (with an 80 percent reflectance coating). The errors are generally much larger for lamps having more-directional intensity distributions (e.g., reflector type lamps), or with integrating spheres having a larger baffle-to-sphere wall area ratio and/or non-uniform reflectance. A method for evaluating and correcting for the angular non-uniformity errors is described in **Annex C**.

7.3.4 Near-Field Absorption. Any objects (such as lamp holders or fins on heat sinking in SSL products) that are very close to the lamp absorb direct light from the lamp and can cause significant errors that cannot be corrected by self-absorption measurement. This effect is called near-field absorption. An example of such a case is shown in **Figure 10**.

In this case, the channel on the luminaire for a linear fluorescent lamp, which is used to slide the lamp holder to adjust lamp length and hold wires, works as a cavity (looks black) and absorbs a significant portion of direct light from the lamp. This would cause serious

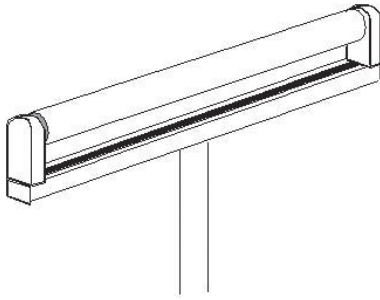


Figure 10. An example of a configuration with serious nearfield absorption. (Image courtesy of Yoshi Ohno and Cameron Miller.)

measurement errors unless the test lamp is compared with a standard lamp of the same type using the same luminaire (strict substitution photometry). There is no way to correct for near-field absorption errors.

To avoid near-field absorption errors, the object (such as lamp holder) should be placed as far from the lamp as possible, the object surfaces coated with high-reflectance coating, and the formation of cavities avoided. The luminaire surfaces facing the lamp should always be kept clean and of high reflectance.

7.3.4 Temperature Effects

7.4.3.1 Photometer Temperature Effect. When a lamp is operated in an integrating sphere photometer, the photometer temperature may change. For example, if a 1000-watt incandescent lamp is measured in a 2-meter integrating sphere several times, the temperature of the photometer can increase by several degrees Celsius. Consequently, the responsivity of the photometer head can change by nearly a half percent, assuming the temperature coefficient of a typical photometer head: 0.1 percent per degree Celsius. These temperature changes may be due to a combination of absorption of radiation by the $V(\lambda)$ filter and photodetector and thermal conduction through the sphere wall material. To avoid this error, a temperature-controlled photometer can be used. If a photometer with no temperature controller or monitoring is used, this error can be reduced by operating a dummy lamp similar to the lamp to be measured in the sphere before measurement starts, to stabilize the integrating sphere photometer within a temperature range for calibration of all lamps.

7.4.3.2 Effect of Heat on the Coating. When a lamp is operated in an integrating sphere photometer, the coating on the sphere wall is heated by radiation from the lamp. In general, the reflectance of commonly used integrating sphere coatings have very small temperature dependences, but the effects of the temperature related change in reflectance are magnified by a factor of $1/(1 - \rho)$ and can affect the sphere responsivity (see **Annex A**). This effect has been difficult to quantify because the coating surface temperature changes very fast (in seconds), but a measurement method is now available [Ohno, Köhler, and Stock]. The effect of heat varies, depending on the coating materials. For the thermal characteristics of the reflectance of a specific sphere coating, the manufacturer should be consulted. This effect will be larger as the size of the sphere is smaller, for the same lamp power. The error due to this effect, if any, can be minimized by strict substitution photometry.

8.0 Uncertainty Evaluation

For general guidance and methods of calculation, please review JCGM 100:2008 and CIE 198:2011. To truly understand the use of an integrating sphere measurement system, a list of all the components of uncertainty in a given measurement system should be evaluated. A typical set of uncertainty components for an integrating sphere measurement system would include:

- Uncertainty of calibration lamp with respect to total luminous or spectral radiant flux
- Aging of calibration lamp
- Absolute scale and stability of lamp electrical control
- Ambient temperature
- Nonlinearity of detector (particularly spectroradiometer)
- Wavelength uncertainty for spectroradiometer
- Stray light of spectroradiometer
- Spectral mismatch correction for photometers
- Self-absorption correction (residual)
- Spatial non-uniformity of sphere (compared to standard lamp)
- Near-field absorption

There will be remaining uncertainty related to the magnitude of the correction made. For self-absorption (**Section 7.3.1**), the residual uncertainty may be evaluated from the stability of the auxiliary lamp during self-absorption correction and the repeatability of opening and closing of the sphere. For spectral mismatch correction (**Section 7.3.2**), the residual uncertainty may be evaluated from the uncertainty of spectral responsivity of the photometer head (for 2π illumination geometry, and usually difficult to measure), and the uncertainty of spectral distribution of test lamps. For sphere angular non-uniformity (**Section 7.3.3**), see, for example, Ohno and Daubach on evaluating the uncertainty contribution. To evaluate thermal effects (**Section 7.3.5**), a technique reported in Ohno, Köhler, and Stock, may be used, wherein an LED auxiliary lamp may be modulated instead of using an external lamp with a mechanical light chopper, while a high-power incandescent lamp is turned on in the sphere to create heat for the sphere and detector, or turned off.

9.0 Maintenance of Integrating Spheres

As an integrating sphere is used, it becomes contaminated, and the sphere responsivity gradually decreases. Contamination not only reduces the average reflectance of the sphere but also degrades the angular non-uniformity of the sphere responsivity. Integrating spheres should be installed in a clean environment to avoid contamination by dust, and the spheres should always be closed when they are not being used. Decrease of average reflectance to some extent is not a problem as long as the sphere is calibrated and the correction factors discussed in this document are verified periodically. The angular non-uniformity of the sphere responsivity affects the measurement accuracy most. Contamination normally occurs more prominently in the lower half of the sphere (particularly around the bottom of the sphere), and the responsivity will be lower where contamination is more prominent. The errors can be significant, particularly for test lamps having uneven distribution of flux upwards and downwards. The contamination by dust or other

airborne particulates can be ameliorated somewhat by 'blowing' out the dust on the lower portion of the sphere. It is recommended that the sphere be re-coated when uneven whiteness of the coating surfaces over the sphere wall is visually conspicuous. Sphere angular non-uniformity can be measured and corrections can be applied. (See **Annex C**).

Some types of sphere coating or paint are subject to a long-term change of color (caused by binder materials), normally losing spectral reflectance in the blue region faster than in the red region; i.e., yellowing. The effects of yellowing are most severe when the integrating sphere with photometer method of measurement is employed. Yellowing leads to a change in spectral mismatch errors. Unless the spectral throughput is measured and corrections adjusted, aged coating exhibiting yellowing should be replaced.

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Annex A – Basic Integrating Sphere Theory

An integrating sphere is a device that angularly integrates the radiant flux generated in (or introduced into) the sphere so that it can be detected with a single photodetector. In the case of lamp measurements, the spatial integration is typically made over the entire (4π) solid angle. The basic theory of an integrating sphere is as follows.

A.1 Flux Transfer in a Sphere

In a hollow sphere, as shown in **Figure A1** and assuming that the sphere surface is lambertian, the illuminance E at point , produced by the luminance L of a surface element having an area a at point , is given according to the inverse-square law and the cosine law as:

$$E = \frac{L \cdot \Delta a \cdot \cos^2 \theta}{AB^2} = \frac{L \cdot \Delta a \cdot \cos^2 \theta}{(2R \cos \theta)^2} = \frac{L \cdot \Delta a}{4R^2}, \quad (A1)$$

where R is the radius of the sphere.

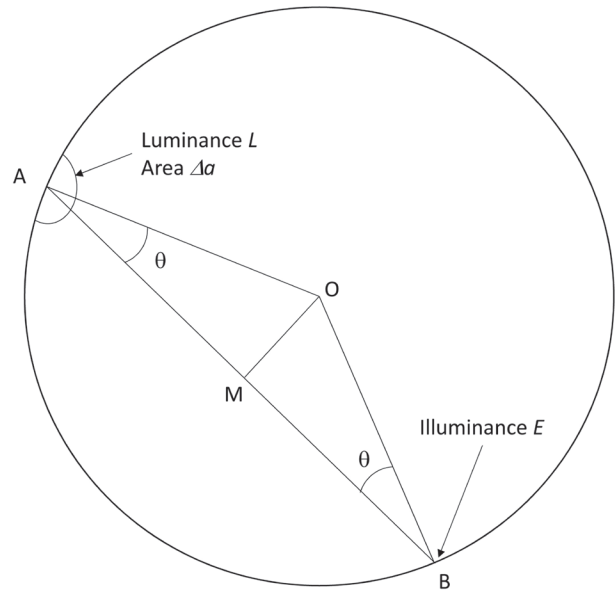


Figure A1. Derivation of the flux transfer theory for a sphere. (Image courtesy of Yoshi Ohno and Cameron Miller.)

Thus, as shown in **Figure A2**, the illuminance E on any part of the sphere wall created by an element of the sphere wall Δa with luminance L is given by:

$$E = \frac{L \cdot \Delta a}{4R^2}, \quad (A2)$$

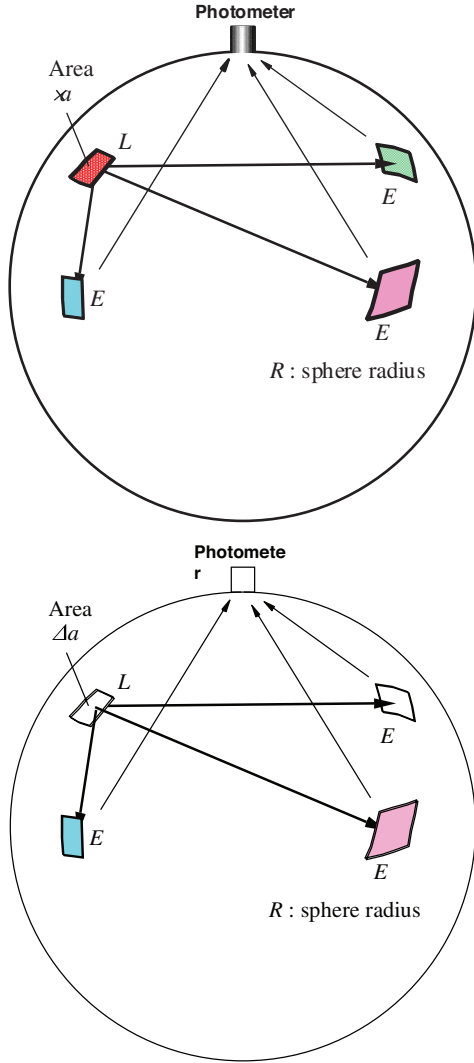


Figure A2. Flux transfer in a sphere. (Image courtesy of Yoshi Ohno and Cameron Miller.)

This means that a given amount of flux incident anywhere on the sphere wall will create an equal illuminance on the detector port. Thus, even without interreflection, one photometer on the sphere wall can theoretically measure the total flux in the sphere. This statement is not strictly valid for a real integrating sphere with less than perfect Lambertian reflectance. However, due to interreflections of light in the sphere, the distribution of reflected light will be close enough to uniformity to approximate the condition assumed in **Equation A2**.

A.2 Measurement of a Lamp

If the basic theory is to be applied for total flux measurement of a lamp, the lamp must be mounted

in the sphere and the detector must receive only the light reflected from the sphere wall. Thus, the direct light from the lamp must be shielded by a baffle, as shown in **Figure A3**. If the reflectance of the sphere wall is not zero, interreflections occur in the sphere. Interreflections help reduce the errors due to non-ideal characteristics of the sphere.

When a lamp with luminous flux Φ_v is operated in a sphere having reflectance ρ , the flux created by interreflections is given by:

$$\Phi_v (\rho + \rho^2 + \rho^3 + \dots) = \Phi_v \cdot \frac{\rho}{1 - \rho}, \quad (\text{A3})$$

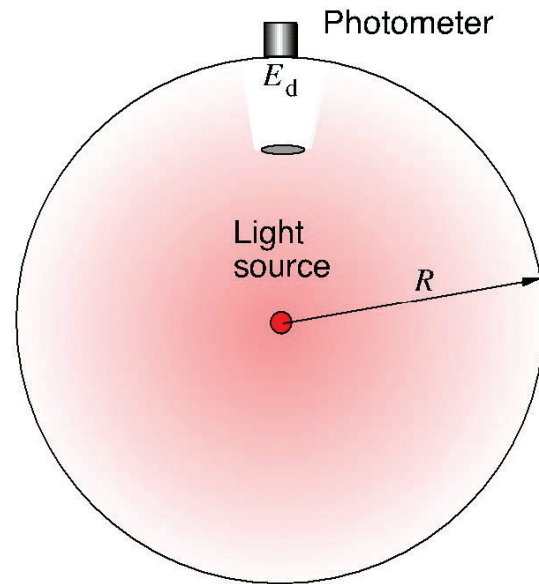


Figure A3. Measurement of a light source. (Image courtesy of Yoshi Ohno and Cameron Miller.)

Then, the illuminance E_d created by all the interreflections is given by:

$$E_d = \frac{\Phi_v \cdot \rho}{1 - \rho} \cdot \frac{1}{4\pi \cdot R^2}, \quad (\text{A4})$$

The photometer head with a perfect cosine response measures E_d as given by **Equation A4**. Then the signal from the photometer head y_{ph} is given by:

$$y_{ph} = E_d \cdot S_v = \frac{\Phi_v \cdot \rho}{4\pi \cdot R^2 (1 - \rho)} \cdot S_v, \quad (\text{A5})$$

where S_v is the illuminance responsivity (A/lx) of the photometer head.

This measurement equation for an integrating sphere photometer indicates that:

The sphere efficacy is strongly dependent on reflectance ρ due to the term $1-\rho$ in the denominator. For example, the signal can be 10 times larger at $\rho = 0.98$ than at $\rho = 0.8$.

The sphere efficiency is very sensitive to a small change of reflectance when it is close to 1, due to the term $(1 - \rho)$ in the denominator. For example, a 0.1 percent change of ρ at $\rho = 0.98$ causes a 5 percent change in the signal.

Annex B – Measurement of the Relative Spectral Throughput of an Integrating Sphere

The relative spectral throughput $T_s(\lambda)$ of an integrating sphere can be obtained by using a spectroradiometer and a reference lamp as shown in **Figure B1**.

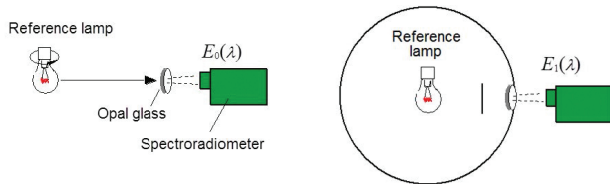


Figure B1: Measurement of the spectral throughput of an integrating sphere. (Image courtesy of Yoshi Ohno and Cameron Miller.)

First measure the relative spectral irradiance $E_n(\lambda)$ of an incandescent reference lamp (clear-bulb type) outside the sphere (e.g., on a photometric bench). If the spectroradiometer measures radiance, use an opal glass to receive light in irradiance mode with cosine response. Calculate $E_n(\lambda)$ as an average of data taken at several different angles of the reference lamp. Then, measure the spectral irradiance $E_n(\lambda)$ of the same reference lamp operated in the sphere using the same combination of the opal glass and the spectroradiometer at the detector port. The opal glass should be mounted flush with the coating surface. The relative spectral throughput $T_s(\lambda)$ of the integrating sphere is given by:

$$T_s(\lambda) = k \cdot \frac{E_i(\lambda)}{E_0(\lambda)}, \quad (\text{B1})$$

where k is a normalization factor.

An example of the spectral throughput and the relative spectral responsivity of an integrating sphere (with approximately a 97% reflectance barium sulfate coating) is shown in **Figure B2**.

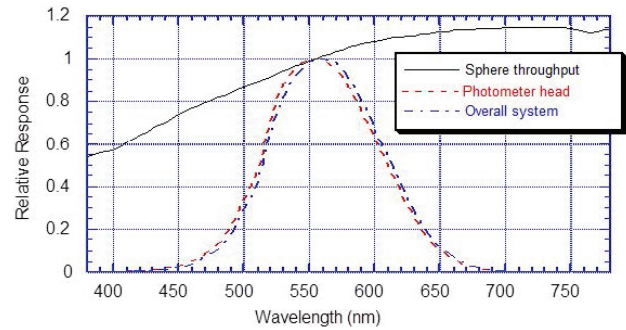


Figure B2. An example of the spectral throughput and the relative spectral responsivity of an integrating sphere photometer. (Image courtesy of Yoshi Ohno and Cameron Miller.)

Annex C – Correction for the Angular Non-uniformity Errors of an Integrating Sphere

As briefly described in **Section 5.3**, the angular non-uniformity of sphere responsivity is expressed in terms of the sphere angular responsivity distribution function (SRDF), $K(\theta, \phi)$. $K(\theta, \phi)$ is defined as the sphere response for the same amount of flux incident on a point (θ, ϕ) of the sphere wall or on a baffle surface, relative to the response at the origin, $K(0,0)$. The SRDF $K(\theta, \phi)$ of an integrating sphere can be obtained by measuring the detector signals while rotating a narrow beam of light inside the sphere as shown in **Figure 4 (Section 5.3)**. If the beam rotation is accomplished by rotating a lamp, the lamp must be insensitive to operating orientation. A schematic curve for a theoretical SRDF is shown in **Figure C1**.

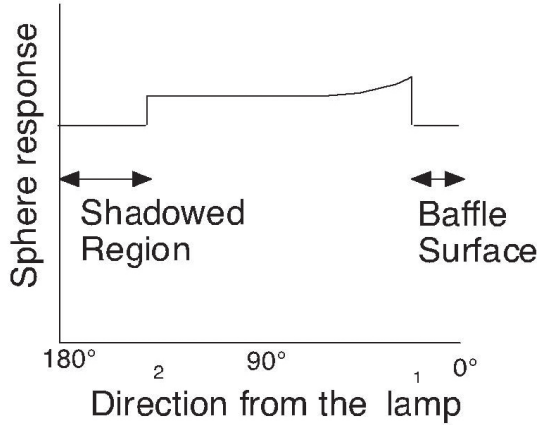


Figure C1. A schematic curve of the theoretical SRDF in the equatorial plane. (Image courtesy of Yoshi Ohno and Cameron Miller.)

The SRDFs of integrating spheres depend not only on the effect of the baffle, but also on the uneven thickness of coating, contamination of its surfaces (particularly the lower hemisphere), the gap between the two hemispheres, and other structures such as the auxiliary lamp and its luminaire. An example of the SRDF of an integrating sphere is shown in **Figure C2**. The errors caused by the sphere angular non-uniformity of an integrating sphere response can be corrected by using the SRDF as follows.

The SRDF measured along the polar coordinates (θ, ϕ) is given as $K(\theta, \phi)$. $K(\theta, \phi)$ is then normalized to $K^*(\theta, \phi)$

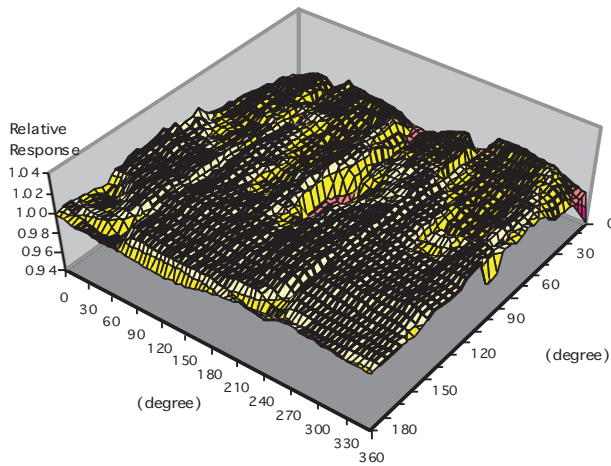


Figure C2. An example of measured SRDF of an integrating sphere (2.5-m sphere, 20-cm baffle, coating reflectance 98 percent). (Image courtesy of Yoshi Ohno and Cameron Miller.)

for the sphere response to an isotropic point source as defined by:

$$K^*(\theta, \phi) = \frac{4\pi K(\theta, \phi)}{\int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} K(\theta, \phi) \sin \theta d\theta d\phi} \quad (\text{C1})$$

The sphere angular non-uniformity correction factor $k_{s,t}$ for the test lamp with respect to an isotropic point source is given by:

$$k_{s,t} = \left[\int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} I^*(\theta, \phi) K^*(\theta, \phi) \sin \theta d\theta d\phi \right]^{-1}, \quad (\text{C2})$$

where $I^*(\theta, \phi)$ is the normalized luminous intensity distribution of the internal source given by:

$$I^*(\theta, \phi) = \frac{I_{rel}(\theta, \phi)}{\int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} I_{rel}(\theta, \phi) \sin \theta d\theta d\phi} \quad (\text{C3})$$

where $I_{rel}(\theta, \phi)$ is the relative luminous intensity distribution of the internal source.

When $k_{s,t}$ is calculated using **Eq. C2**, only a relative intensity distribution is necessary, and its accuracy is not critical. For example, the data for a group of lamps of the same type can be represented by one lamp.

Annex D – Alternative Geometries and Calibration Methods

Integrating spheres used for lamp measurements normally employ either 2π geometry, where the lamp under test is mounted at the opening on the sphere wall, or 4π geometry, where the lamp under test is mounted in the center of the sphere. Other geometries, including integrating hemispheres, are also used. A typical configuration of an integrating hemisphere is shown in **Figure D1**.

This system has similar advantages to the 2π sphere geometry in that it is easy to mount the lamp and that the light absorbing, non-emitting surfaces of the test lamp are outside the hemisphere and do not affect self-absorption. Compared to the 2π integrating sphere, the integrating hemisphere has an advantage in that the test lamp hardware does not affect self-absorption of the sphere, as well as convenience in mounting lamp. The flat cross-section surface of the hemisphere must

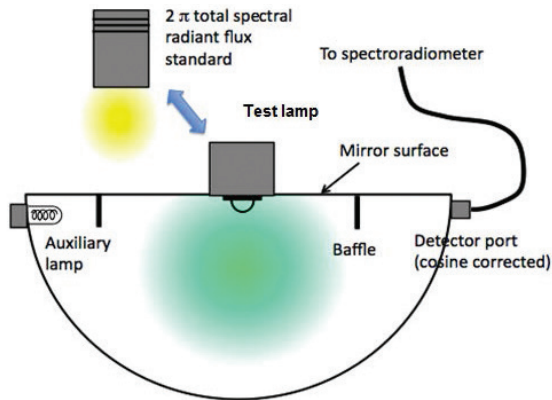


Figure D1. Typical configuration of an integrating hemisphere. (Image courtesy of Yoshi Ohno and Cameron Miller.)

be a high-reflectance specular reflector. In principle, this system is suitable for measuring 2π sources (like some solid-state lamps) and should be calibrated with a 2π standard lamp. It is not recommended to calibrate the system against a 4π standard lamp and measure 2π sources or vice versa.

The orientation of the hemisphere sphere can be in any direction. In other words, the figure can be taken as side-view or top-view. However, note that incandescent standard lamps are sensitive to operating position and need to be calibrated in the operating position to be used.



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